

	Case Name: Apartment Building (6)	Sector	Construction (residential building)
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources
			Schedule with costs
		Risk Analysis	Random simulation
Submitted by	Matilde Casado and Margarida Antunes		One of nine std. scenarios
Date	2015		User defined distributions
File Name	C2015-05 Apartment Building (6)	Project Control	Automatic tracking
			Tracking based on user input

1. Project description

Project authenticity

The project includes roofing, ventilation, insulation, electrical, plumbing, flooring, plastering, kitchen installation, and finishing works, ensuring a modern and fully functional residential space.

The project consists of activity, resource and cost data that were obtained directly from the actual project owner.

2. Project properties

2.1. Baseline Schedule

General		Network topology	
# Activities	64	Serial/Parallel (SP)	22%
Planned Duration (PD)	120 days*	Activity Distribution (AD)	41%
Budget At Completion (BAC)	2 524 765.2 €	Length of Arcs (LA)	0%
Renewable Resources	-	Topological Float (TF)	51%
Consumable Resources	-		

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity				Time sensitivity		
	avg [%]	std dev [%]	skew [-]		avg [%]	std dev [%]	skew [-]
CRI-r	11.9	11.6	-3.0	CI	21.1	33.9	-1.2
CRI-rho	11.6	11.2	-3.2	SI	36.3	34.1	-0.8
CRI-tau	7.0	7.7	-3.6	SSI	6.0	9.1	-1.1
				CRI-r	12.6	9.3	-0.8
				CRI-rho	12.7	9.0	-0.8
				CRI-tau	8.2	5.2	-0.7
	Resource sensitivity						
	avg [%]	std dev [%]	skew [-]				
CRI-r	N/A	N/A	N/A				
CRI-rho	N/A	N/A	N/A				
CRI-tau	N/A	N/A	N/A				

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy			Simulated EAC accuracy		
method - PF	MAPE [%]	MPE [%]	method (PF)	MAPE [%]	MPE [%]
PV - 1	12.7	-10.8	1	1.5	0.2
PV - SPI	12.8	3.1	CPI	2.6	0.3
PV - SCI	14.6	2.7	SPI	10.1	10.0
ED - 1	19.6	-19.6	SPI(t)	10.8	10.8
ED - SPI	12.8	3.1	SCI	10.1	9.9
ED - SCI	13.7	2.8	SCI(t)	10.9	10.7
ES - 1	10.2	-6.2	0.8 CPI + 0.2 SPI	3.6	3.2
ES - SPI(t)	18.5	15.1	0.8 CPI + 0.2 SPI(t)	4.2	3.6
ES - SCI(t)	19.3	14.8			

According to the MAPE values¹ the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 5 tracking periods with a length of approximately 1 month. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

¹ The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	-13033.2	-118535.9	-93	1.0	0.9	0.9	1.0
std dev	24133.8	95608.4	66.1	0	0	0	0
final	-38911.0	0	-88	1.0	1.0	1.0	1.0

2.3.3.2. Time forecasting

PD	120 days	Real Duration	131 days	Late	9.2%
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EAC(t)	Real Accuracy			
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	125.6	0.9	0.8	0.8
PV - SPI	126.2	1.3	0.7	0.7
PV - SCI	127.0	1.6	0.6	0.6
ED - 1	128.0	1.6	0.5	0.5
ED - SPI	128.8	1.3	0.3	0.3
ED - SCI	129.4	0.9	0.2	0.2
ES - 1	130.0	0.7	0.2	0.2
ES - SPI(t)	130.6	0.9	0.1	0.1
ES - SCI(t)	131.2	1.3	0.0	0.0

2.3.3.3. Cost forecasting

BAC	2 524 765.2 €	Real Cost	€ 2,563,676	Over Budget	1.5%
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EAC	Real Accuracy			
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	2537798.2	24133.8	0.2	0.2
CPI	2536815.5	26695.9	0.2	0.2
SPI	2589298.7	40138.3	0.2	-0.2
SPI(t)	2617900.8	71868.5	0.4	-0.4
SCI	2588252.0	40981.1	0.2	-0.2
SCI(t)	2616905.0	72493.7	0.4	-0.4
0.8 CPI + 0.2 SPI	2546588.2	20614.6	0.1	0.1
0.8 CPI + 0.2 SPI(t)	2551372.8	21341.4	0.1	0.1